

Firm, Market, and Equilibrium Responses to Behavioral Frictions

Behavioral Labor – Week 8

Teaching deck draft

Central question

- Once worker-level behavioral wedges exist, how do firms and markets respond?
- Week 8 follows the course sequence:
 - Weeks 2–4: worker wedges in preferences, beliefs, attention, and complexity.
 - Weeks 5–6: workplace and allocation environments.
 - Week 7: measurement and identification.
- Today: what survives after firm strategy, search, sorting, and competition?

Worker-level wedges are not enough

$$Y_{im} = g(\theta_i, R_f(\theta_i, \mathcal{M}_m), \mathcal{M}_m)$$

- θ_i : worker-level wedge such as inertia, misperception, or low salience.
- $R_f(\cdot)$: firm response through menus, defaults, messages, fees, wages, or recruiting.
- $\mathcal{M}_m$: market environment: competition, transparency, switching costs, search frictions, heterogeneity, market power.
- Observed choices are already equilibrium-shaped.

Taxonomy of firm responses

Response	Firm action	Empirical objects
Exploit	design fees, defaults, or menus around inertia and confusion	shrouded attributes, switching, fee sensitivity, passive choice
Accommodate	simplify choices or make valuable actions easier	active choice, clearer menus, reduced mistakes, better matching
Insure	use defaults, restricted menus, or benefits to protect workers	contribution floors, benefit design, risk exposure, retention
Screen / sort	use messages, contracts, or menus to separate worker types	applicant pools, wage bids, plan choice by type, match quality

The same practice can insure some workers and exploit or segment others.

Employer-sponsored benefits as market response

- Benefits are compensation, but they are delivered through menus, defaults, deadlines, fees, and employer interfaces.
- Worker margin: enrollment, contribution, plan choice, switching.
- Firm or intermediary margin: menu design, default choice, disclosure, sales effort.
- Market margin: fee competition, demand elasticity, sorting, welfare.
- Core anchors: Bernheim–Fradkin–Popov; Handel–Kolstad; Duarte–Hastings; Hastings–Hortacsu–Syverson.

401(k) defaults and welfare

Question	Bernheim–Fradkin–Popov logic
Identifying variation	workers face different retirement-plan defaults and menu environments
Observed margin	participation, contribution rates, asset allocation, default persistence
Worker-level effect	inertia, procrastination, passive choice
Firm response margin	employer default and menu design
Welfare issue	default-induced saving is not automatically welfare-improving

Large default effects are evidence of a behavioral wedge; they are not a complete welfare statement.

Plan choice, inertia, and transparency

Paper domain	Observed margin	Equilibrium lesson
Health insurance for “humans”	plan choice, expected costs, risk exposure	information frictions create welfare losses inside employer-mediated menus
Mexico social security disclosure	switching, fee sensitivity, account choice	transparency can raise demand elasticity, but firms may adjust marketing and sales effort
Sales force and competition	demand, fees, switching, plan shares	competition works through sales-force strategy, disclosure, and consumer sophistication

The question is not only whether workers make mistakes. It is whether markets discipline or monetize those mistakes.

Distorted outside-option beliefs

$$\tilde{w}_i^{out} = w_i^{out} - \mu_i$$

- Jager–Roth–Roussille–Schoefer: workers may misperceive outside options.
- Observed margins: beliefs, search, bargaining, mobility, wage response.
- If $\mu_i > 0$, workers perceive outside options as worse than they are.
- Firms may retain workers at lower wages when perceived mobility is low.
- The belief object becomes a segmentation and market-power object.

Search equilibrium and segmentation

- Week 3 question: do beliefs affect worker search?
- Week 8 question: do belief distortions change market allocation?
- Menzio's stubborn-beliefs logic asks when market experience fails to correct beliefs.
- Search frictions, weak wage transparency, and low mobility can preserve distorted beliefs.
- Frontier: behavioral monopsony, segmentation, and wage markdowns under distorted outside options.

Firm messages, sorting, and search direction

- Horton–Kuchler–Stanton: employer messages can affect applications, wage bids, and sorting.
- Communication may reduce uncertainty and improve match quality.
- Communication may also screen worker types or steer search strategically.
- The empirical distinction:
 - worker response to messages;
 - firm choice of messages;
 - market effect on matching and wage dispersion.

When do wedges survive?

Condition	Attenuation	Survival or amplification
Competition	clear total compensation and better menus	competition on salient attributes while shrouding hidden costs
Transparency	workers compare wages, benefits, fees, outside options	attributes are opaque, delayed, or broker mediated
Switching costs	workers can change plans or jobs cheaply	default persistence and slow job or plan switching
Search frictions	offers and outside options are learned quickly	low search and segmented information networks
Heterogeneity	sophisticated workers discipline markets	firms target less sophisticated workers
Labor market power	workers can credibly leave	firms retain workers with pessimistic beliefs or low mobility

Identification map

Block	Identifying variation	First object identified
Retirement defaults	default exposure and menu variation	worker default response under a given menu
Health-plan choice Retirement account markets	information and plan-choice environment disclosure, sales force, competition	plan-choice mistakes and welfare losses demand elasticity, switching, fee sensitivity
Outside-option beliefs	elicited beliefs and objective outside-option measures	belief distortion and worker search response
Labor-market messages	message or platform variation	application, bid, and sorting response

Equilibrium interpretation needs extra evidence on firm response and market outcomes.

Partial versus equilibrium welfare

- Attenuation is not always correction.
- Amplification is not always manipulation.
- A partial effect can disappear or reverse after firms redesign menus, fees, wages, or messages.
- Revealed preference is weaker when choice architecture is endogenous.
- Welfare requires a benchmark: full information, low hassle, sophisticated choice, structural utility, or planner target.

- Reproduce: retirement-default exercise anchored on Bernheim–Fradkin–Popov.
- Diagnose: menu design, default quality, passive choice, and welfare distance.
- Transfer: outside-option beliefs and segmentation anchored on Jager–Roth–Roussille–Schoefer.
- Optional extension: disclosure, sales effort, demand elasticity, and switching costs from Duarte–Hastings or Hastings–Hortacsu–Syverson.

- Week 9 asks how policy should be designed once firms and markets respond.
- Defaults, disclosure, simplification, and nudges are not implemented in a vacuum.
- Policy can change worker behavior and firm strategy at the same time.
- The policy target must name the worker wedge, firm response, market condition, and welfare benchmark.

- Worker-level behavioral wedges are not observed market outcomes.
- Firms can exploit, accommodate, insure, screen, sort, and segment.
- Competition helps only under the right transparency, switching, and search conditions.
- The empirical frontier links worker behavior to firm response and market outcomes.
- Welfare and policy require equilibrium interpretation, not only strong reduced forms.